

Chun-Wei Chen

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EDUCATION

National Tsing Hua University

M.S. Student, College of Semiconductor Research

Sep. 2025 – Jun. 2027 (Expected)

Hsinchu, Taiwan

- Advisor: Prof. Ting-Chi Wang
- Relevant Coursework: VLSI Physical Design Automation, Machine Learning for EDA, Computer Architecture, Essentials of Intelligent System Design Flow

National Central University

B.S. in Electrical Engineering | GPA: 4.0/4.3

Sep. 2021 – Jun. 2025

Taoyuan, Taiwan

EXPERIENCE

Synopsys – AI-Assisted Analog APR Configuration Authoring

Jul. 2026 – Aug. 2026

R&D Intern, Custom Compiler Team

Hsinchu, Taiwan

- Built a **deterministic linter** for process-dependent analog place-and-route YAML, with ~50 schema checks and ~20 cross-key semantic rules in a **declarative YAML DSL**, keeping the engine extensible to new production parameters.
- Added an explanation layer over an internal **RAG** service that cites the governing spec section for each finding.
- Shipped as a **CLI** plus **Cursor** agent skills; validated on a 26-fixture harness spanning diverse finding types, correctly identifying and explaining every planted finding, and cut retrieval calls ~80% by skipping them when validator output is already self-sufficient.

NSTC Project – Thermal-Aware Global Routing for SoC Reliability

Sep. 2025 – Present

Advisor: Prof. Ting-Chi Wang

Hsinchu, Taiwan

- Developing a **thermal-aware global router** for SoC physical design by incorporating hotspot distribution into multi-objective routing cost optimization, including overflow, wirelength, via count, and thermal hotspot avoidance.
- Building a **PyTorch-based thermal map predictor** using a U-Net with attention architecture to enable high-speed thermal inference during global routing.
- Generating and analyzing thermal simulation data using **Cadence Voltus** and **OpenROAD**.

2024 ICCAD CAD Contest – Chip-Level Global Router

Jun. 2024 – Sep. 2024

Advisor: Prof. Yu-Guang Chen

Tainan, Taiwan

- Received an **Honorable Mention** in Problem D of the 2024 ICCAD CAD Contest.
- Developed a chip-level global routing algorithm jointly optimizing **wirelength, channel overflow, via count, and edge pin density**.
- Implemented an **Obstacle-Avoiding Rectilinear Spanning Tree** algorithm to minimize wirelength while navigating around placement obstacles.
- Applied a modified A* routing algorithm to reduce via count and edge pin density after initial routing tree generation.

PROJECTS

Detailed Placement Optimizer | C++, VLSI Physical Design Automation

2025

- Built a standard-cell detailed placement optimizer supporting **Global Swap, Vertical Swap, Local Reordering, and Single-Segment Clustering**.
- Designed an incremental cost evaluation method using partial HPWL and per-net minimum/maximum coordinate caches.
- Implemented move-to-empty and mutual-overlap legality checks, with placement candidates retrieved using bin and segment indices.

Python-Based SPICE Circuit Simulator | Python, Numerical Methods

2024

- Developed a Python-based SPICE circuit simulator with a netlist parser supporting **AC, DC, transient analyses**, and **subcircuit statements**.
- Applied **Newton's Method, Backward Euler, BDF2, and Modified Nodal Analysis** for nonlinear and transient circuit simulation.
- Successfully simulated circuits containing up to **234 MOSFETs**, with an average error of approximately **1–2%** compared with Ngspice.

SKILLS

Programming Languages: C, C++, Python, Tcl, PyTorch

Tools: Git, Linux, JSON Schema

EDA Tools: Cadence Voltus, Cadence Innovus, Cadence Virtuoso, Cadence Allegro Package Designer, Synopsys Custom Compiler, Synopsys HSPICE, OpenROAD

Languages: Mandarin Chinese (Native), English (TOEIC 895)